

continuation

Continuation Document

Revision: 96 **Last modified:** 2026-06-02T00:59:25Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
9327930	Foundation wave 30: mark 3 items Completed (HXC-1302/1299/1291) — wireguard spot-drain teardown / discovery 15-tier registry / device.Probe real host detection. Mutation spot-checks confirm bites. Registry 202->205 Completed / 434 Queued.
e4f0b03	Foundation wave 30: 3 disjoint streams (all approved no-fix; mutation spot-checks bite). HXC-1302 wireguard preemption-safe teardown (pkg/wireguard/teardown.go: MeshCoordinator.TearDownPeers Stop+RemovePeer-all idempotent + SpotDrainHook(mc) cloudspot.Hooks.StopAdmission; tests w/ real curve25519 keys: add N -> teardown -> ListPeers==0, idempotent, FakeSource+Handler.Run drain removes peers; mutation skip-RemovePeer fails). HXC-1299 discovery 15-tier registry (pkg/discovery/tier_registry.go: TierRegistry.Register+SelectByTier(minTier) rank>=min via tierdef.Default() T1-T15 ordering, ordered exact-set; mutation >=<

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1e88aec	fails TestSelectByTier_T8 got T1/T5 want T8/T12/T15). HXC-1291 device.Probe (pkg/device/probe.go + probe_{darwin,linux,other}.go: real Arch(GOARCH)/CPUCores(NumCPU)/MemoryBytes(sysctl hw.memsize / syscall.Sysinfo / MemStats fallback) + REAL os.Hostname-based ID, honest 0 for GPU/NPU/power; tests assert vs runtime oracle + real sysctl). ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; B mutation bite + A/C reviewer-verified. Foundation wave 29: mark 3 items Completed (HXC-1298/1296/1310) — handheld scheduling policy / device<->pb mapping (honest lossy) / per-tier security model. Mutation spot-checks confirm bites. Registry 199->202 Completed (crossed 200) / 437 Queued.
441c7fe	Foundation wave 29: 3 disjoint streams (all approved; A minor dead-helper + comment-lie fixed by me). HXC-1298 power/battery/thermal-aware handheld scheduling policy (pkg/scheduler/handheld.go HandheldPolicy Plugin reusing edgeaware battery/thermal/charging helpers: filter low-battery/over-thermal/unavailable/discharging-below-floor, score by battery/thermal/charging; absent-label assume-safe; 22 tests; mutation pct< -> pct> fails LowBatteryRejected). HXC-1296 device.Descriptor<->pb.Node round-trip mapping (NEW pkg/devicemap: DescriptorToNode/NodeToDescriptor over MAPPABLE fields; HONEST lossy – round-trip identity over mapped subset + explicit test that device-only fields NPU/power/battery/trust are LOST (no false-recovery bluff); no import cycle; 7 tests). HXC-1310 per-tier security model enforcement (NEW pkg/tiersec reusing sandbox.Capability+edge+tierdef: ProfileForTier + Enforce(tier,op)->Decision over capability/data-class(inclusive)/network-egress CIDR allowlist via stdlib net; low-tier denied / high-tier allowed both directions; 24 tests; mutation data-class > -> >= fails inclusive-boundary). ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; A+C

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634aa09	mutations bite, B honest-loss reviewer-verified. Foundation wave 28: mark 3 items Completed (HXC-1311/1301/1303) — T1-T15 tier definitions / cloudspot PreemptionWatcher (verified) / inference Backend+Router (verified, dual-layer mutation). Registry 196->199 Completed / 440 Queued. Foundation wave 28: HXC-1311 implemented + HXC-1301/1303 verified-already-done. HXC-1311 T1-T15 tier definitions (NEW pkg/tierdef: //go:embed tierdef.yaml with all 15 ascending tiers + min_requirements; LoadTiers parse+validate (rejects bad schema_version, missing/dup/out-of-order/wrong-count), Default(); TierDef.Meets(DeviceCaps) inclusive >= per-dimension (cpu/mem/gpu/vram/net); 73 test cases assert exact parsed T1/T15 values + per-dimension below-boundary false + exact-boundary true; mutation < -> <= fails TestMeets_ExactBoundary_True).
d4829d1	HXC-1301 pkg/cloudspot PreemptionWatcher VERIFIED already complete (injectable PreemptionSource + 3-step graceful drain StopAdmission/Checkpoint/Deregister + deadline ForceStop + FakeSource + 9 mutation-killing tests). HXC-1303 pkg/inference provider-agnostic Backend interface + Router registry VERIFIED already complete (capability-aware Candidates/Route fallback + ReferenceBackend labeled output + ErrUnsupported; DUAL-LAYER capability correctness confirmed by me via mutation: neutering BOTH the Candidates filter AND the backend model gate fails TestRouterPicksCapabilityMatch/UnsupportedErrors – defense-in-depth, not a bluff). ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; 1311 boundary mutation + 1303 dual-layer mutation bite.
5745f9f	Foundation wave 27: mark 2 items Completed (HXC-1237/1278) — swim prod+sim transport parity (real protocol over deterministic SimNetwork) / property-based fuzzing (4 Go fuzz targets). Mutation spot-checks confirm bites. Registry 194->196 Completed / 443 Queued.

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656bec1	Foundation wave 27: 2 disjoint streams (both approved no-fix; mutation spot-checks bite). HXC-1237 swim prod+sim transport parity (pkg/swim: minimal additive Config.Transport injection seam in NewProtocol – nil => real UDP NewTransport as before, non-nil => injected MessageTransport; prod behavior unchanged. NEW simparity_test.go runs the REAL swim membership protocol over the deterministic SimNetwork: 3-node full-mesh converges (all see StateAlive, 12 msgs), 4-node determinism (same seed identical), seam-load-bearing test asserts sim addr 127.0.0.1:18200. mutation: ignore cfg.Transport -> node sees only itself, convergence FAILS). HXC-1278 property-based + coverage-guided fuzzing (4 native Go FuzzXxx targets, NEW files only): FuzzParseCoverProfile (no-panic + Covered<=Total + Aggregate in [0,100]), FuzzParseTraceParent (no-panic + String() round-trip stable + valid hex ids), FuzzApply (checkpoint_merge no-panic on arbitrary checkpoint bytes via real checksum/gob path), FuzzCRDTCheckpointRoundTrip (NewCRDTCheckpoint->RestoreCRDT Fingerprint-stable + corrupt-bytes no-panic). All assert falsifiable invariants on non-empty seed corpora, run under plain go test (no -fuzz needed); mutation: break TraceParent.String() round-trip FAILS. ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; 2 mutation spot-checks bite.
5707d6b	Foundation wave 26: mark 3 items Completed (HXC-1268/1265/1236) — helix-snapshot CLI / capability sandbox+resource-limits+audit / HelixNetwork trait (prod+sim). Mutation spot-checks + real-binary verified. Registry 191->194 Completed / 445 Queued.
7c41c2f	Foundation wave 26: 3 disjoint streams (all approved; A minor comment-lie fixed by me). HXC-1268 standalone cmd/helix-snapshot CLI (run() dispatch + create/restore/compare/list/delete wiring real snapshot.Manager via -dir; 11 tests w/ real temp-dir fs assertions + exit codes; verified by me running the REAL binary: create writes .golden, list shows it, compare rc=0,

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	delete removes it). HXC-1265 capability sandbox + REAL resource-limit enforcement + audit (NEW pkg/sandbox: Capability deny-by-default grant model + Guard.Check (deny ungranted) + atomic.Int64 cumulative MaxOps/MaxBytes/MaxDuration enforcement – the genuine gap vs pkg/wasm caps – + MemAudit allow/deny log; in-process model, OS syscall isolation honestly out-of-scope not faked; mutation neuter-Has -> ungranted allowed FAILS). HXC-1236 HelixNetwork trait prod+sim (NEW pkg/helixnet: HelixNetwork interface + ProdNetwork real in-process channel fabric (bytes genuinely move A->B, deliver errors on unknown/closed dst) + SimNetwork deterministic via dst.Engine; zero pkg/swim edits – that’s 1237; mutation drop-delivery -> send/unknown-addr tests FAIL). ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green (only the known internal/ advisory load-flake, passes isolated), dataplane ok, tmux/screen clean; 3 mutation spot-checks bite + helix-snapshot real-binary verified.

§2: Environment Snapshot

Property	Value
Branch	main
Commit	9327930
Timestamp	2026-06-02T00:59:25Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919		✓ Done

HXC	Title	Status
	htmux, messaging, GPU, WASM, storage, build manifest	
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

Run chaos tests

```
go test -race -count=1 ./test/chaos
```